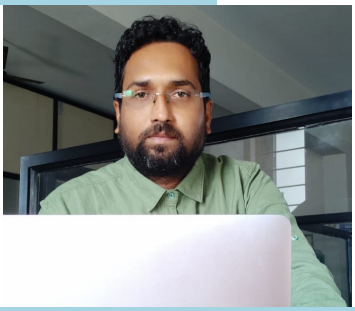
# Panchanand Jha *(B.E., MTech., Ph. D)*




Software Development Engineering in Test, Zucitech Software Pvt. Ltd. Bengaluru, India

+91 8309367652

jv15700[at]gmail.com

https://jaguar1617.github.io/panchanandjha/


## Overview

Researcher with 7+ years of experience in Robotics, data science, machine learning, AI, and IoT. Holding a Master of Technology and a PhD in robotics and machine learning, I am committed to staying updated with the latest advancements in AI/ML and Robotics. I possess sound knowledge in Python, JavaScript, TypeScript, Swift, R-language, Tableau, Tableau Prep, Node-red, Heroku, Django, Streamlit, Flask, and MySQL. With my expertise and passion, I strive to solve real-time challenges and deliver outstanding results in every project.

## Educations and certifications

***Ph.D.*, Industrial Design, (8.99/10 CGPA)** — **07/2011 – 01/2016**
National Institute of Technology Rourkela (India)

Thesis: Inverse Kinematics Analysis of Robot Manipulators (Application of AI/ML and soft-computing Techniques)

***MTech.*, Mechanical Engineering, (8.63/10 CGPA),** — **07/2007 – 06/2009**
National Institute of Technology Rourkela (India)

Thesis: Artificial intelligence neural network application for the prediction of kinematics of robot manipulator.

***B.Tech*, Industrial and Production Engineering, (7.02/10 CGPA),** — **07/2003 – 06/2007**
Central University Guru Ghasi Das Bilaspur, (India)

IBM Certification on *Machine learning with python,* *(https://courses.360digitmg.skillsnetwork.site/certificates/01e6e668e9d84673a2c15dbf92d244ec)* — **2021**

Data Science *certifications from 360digitmg, (Certificate No. 360DTMGIN9116, https://360digitmg.com/aispry-ind-verify)* — **2021**

360digitmg certification on *Tableau*, Tableau Prep. (Certificate No. 360DTMGIN7513, https://360digitmg.com/aispry-ind-verify) — **2021**

## International Research fellowship

- ***Researcher*** *at Department of Computer Engineering and Systems Science*, University of Pavia, Italy, 2017
- ***Researcher*** *at Department of Robotics and Cybernetic*, Slovak Technical University in Bratislava, Slovak Republic, 2014

## Interests

- Data Science
- Machine Learning
- Mobile Robotics
- Industrial Robotics
- Computer vision
- Artificial Intelligence
- Engineering Optimization
- Ergonomics in the Automotive Design
- IoT

## Profiles

- GitHub https://github.com/JAGUAR1617
- Google Scholar https://scholar.google.com/citations?user=DRik19cAAAAJ&hl=en
- Orcid https://orcid.org/0000-0002-7382-5463
- Tableau https://public.tableau.com/app/profile/p.jha

## Technical Skills

- Active user of Python for AI/ML development
- Frequent user of JavaScript/TypeScript with Webdriverio for BDD and TDD
- Active user of Python with Robot Framework for Automation Engineering
- Actively using GitLab, Agile software development (JIRA)
- Active user of GitLab CI/CD pipeline
- Experience in API Test Automation using **python, robot framework with marshmallow library**
- Good knowledge of automation using **Swift** and **XCUIT Test**
- Personal use of HTML, CSS for front-end development
- R-language for data science and machine learning
- Processing.js, Node-Red for IoT based real time projects.
- Pytorch, TensorFlow, OpenCV for image classifications

## Projects

- **Defect Checker:** A pre-trained and fine-tuned version of sentence-transformers/nli-mpnet-base-v2 model on custom dataset for sentence similarity and classification. (Zucitech Software Solutions Pvt. Ltd. Bengaluru, India, 2022).
- **Currently engaged on Disney+ project.**
- Forecasting the prices of agriculture commodities, (Innodatics Inc. USA, 2021), (https://www.linkedin.com/posts/panchanand-jha-7767674b_machine-learning-project-iii-activity-6965160084297408512-J4tb?utm_source=linkedin_share&utm_medium=member_desktop_web)
- Named Entity recognition and feature extraction from medical journals, (Innodatics Inc. USA, 2021), (https://www.linkedin.com/posts/panchanand-jha-7767674b_data-science-project-intern-ii-activity-6965159882337497088-nIwo?utm_source=linkedin_share&utm_medium=member_desktop_web)
- Mortality risk analysis with python and Tableau, (Innodatics Inc. USA, 2021), (https://www.linkedin.com/posts/panchanand-jha-7767674b_data-science-project-intern-i-activity-6965159638925225985-zyaj?utm_source=linkedin_share&utm_medium=member_desktop_web)

## Web deployments

- https://share.streamlit.io/jaguar1617/diabetes-ensemble-classifier/main/ demo.py
- https://jobpredictionapp.herokuapp.com/
- https://share.streamlit.io/jaguar1617/image-processing-web-app/main/ main.py

## Research Publications

**Patents:**

- 29-00-Devices and equipment against fire hazards for accident prevention and for rescue (Design of Rescue Robot), Application No. 360061-001, Indian Patent. **(Granted)**
- The Methodology of Frequency Characterization for More Complex Geometries in a Diesel Engine, Indian Patent, Application No. 202231006305, Indian Patent. **(Published)**
- System and method for analysis of sentiment classification of movie review using GA, NeuroGA, "Application No. 202241047604, Indian Patent **(Published)**

***Scopus/SCI Articles:***

- **Panchanand Jha,** "A hybrid ensemble learning model for evaluating the surface roughness of AZ91 alloy during end milling operation", Surface Review and Letters, 2022, https://doi.org/10.1142/S0218625X23400012

- Din Bandhu, G. Praveen Kumar Yadav, Kumar Abhishek, Jay J. Vora, **Panchanand Jha**, Kuldeep K Saxena, Tushar R Dandekar, Soni Kumari, "Elucidating the effect of Metal-Cored Filler Wire for Butt-welded ASME SA387-Gr.11-Cl.2 steel during Gas Metal Arc Welding Process", Journal of Adehesion Science and Technology, Taylor and Francis, June, 2023, **(Accepted)**

- **Panchanand Jha** and B B Biswal, "Human posture key points extraction and its implementation on upper extremities of humanoid robot", Journal of intelligent and robotic system, (2022), (**Under Review)**

- Panchanand Jha and B B Biswal, “An efficient quantised deep convolution neural network for the prediction of COVID-19”, Applied Artificial Intelligence (2021), **Under Review**

- Panchanand Jha and B B Biswal, “A mathematical approach for creative graphics design”, Journal of graphic engineering and design; 2020; Vol 11(1); Pages 37-46

- Adithya Kolakoti, **Panchanand Jha**, Prakasa Rao Mosa, Manohar mahapatro, Tulasi Ganesh Kotaru, “Optimization and Modelling of Mahua Oil Biodiesel using RSM and Genetic algorithm technique”, Journal of Vibroengineering, Mathematical Models in Engineering, Vol. 6, Issue 2, 2020. https://doi.org/10.21595/mme.2020.21357**.**

- Abinash Tripathy, **Panchanand Jha and** Gandi Satyanarayana, "USE OF INDUCTIVE SEMI SUPERVISED APPROACH-BASED SENTIMENT ANALYSIS FOR MOVIE REVIEWS", IJRAR - International Journal of Research and Analytical Reviews (IJRAR), E-ISSN 2348-1269, P- ISSN 2349-5138, Volume.6, Issue 2, Page No pp.110-118, May 2019, Available at : http://www.ijrar.org/IJRAR19K9414.pdf. (UGC CARE Journal)

- Abinash Tripathy, Ch. Chakradhar Rao, **Panchanand Jha** and Gandi Satyanarayana, “Aspect based approach for document level sentiment classification of movie reviews”, Journal of Emerging Technologies and Innovative Research (JETIR), Volume 6, Issue 3, Pp 85-92, March 2019, ISSN-2349-5162. Available at: https://www.jetir.org/papers/JETIR1903N08.pdf (UGC CARE Journal)

- Abinash Tripathy, **Panchananda Jha**, Rugada Vaikunta Rao, Ch.Srinivasu, "SENTIMENT CLUSTERING OF MOVIE REVIEW DATA USINGUNSUPERVISED MACHINE LEARNING ALGORITHM", International Journal of Creative Research Thoughts (IJCRT), ISSN:2320-2882, Volume.6, Issue 1, pp.745-753, March 2018, Available at :http://www.ijcrt.org/papers/IJCRT1134557.pdf

- Abinash Tripathy, Panchananda Jha, Rugada Vaikunta Rao, Ch.Srinivasu, “ Neuro-Fuzzy Clustering Approach for Sentiment Analysis", Journal of applied science and computations, Volume 6- Issue 4- April- 2019-2, DOI:16.10089.JASC.2018.V6I4.453459.0500101370

- Panchanand Jha and B B Biswal, “A hand haar cascade using OpenCV based gesture control mobile robot for navigation”, Defense Technology, Elsevier, (2021), **Under Review**.

- Panchanand Jha and B B Biswal,“Intelligent Computation and Kinematics of 4-DOF SCARA Manipulator using ANN and ANFIS”, Proceedings of Springer-Verlag Berlin Heidelberg, LNCS 8298.

- Panchanand Jha, Bibhuti B. Biswal, and Om Prakash Sahu, "Intelligent Computation of Inverse Kinematics of a 5-dof Manipulator Using MLPNN," Proceedings of Springer-Verlag Berlin Heidelberg, LNAI 8717, pp. 243–250, 2014.

- Om Prakash Sahu, Bibhuti Bhusan Biswal, Saptarshi Mukharjee, and **Panchanand Jha**, "Multiple Sensor Integrated Robotic End-effectors for Assembly," Procedia Technology, Elsevier, 14, 100-107, 2014.

- Bunil Kumar Balabantaray, **Panchanand Jha** and B B Biswal,“Application of Edge Detection Algorithm for Vision Guided Robotics Assembly System”, Proceeding of SCIEI.

- Panchanand Jha et, al., “Generation of Optimized Robotic Assembly Sequence using Immune Based Technique”, Proceedings of the ASME.

- Nethala Raju, Panchanand Jha and T Ganapathy, "Machining Parameters Optimization on Parallel Kinematic Milling Machine using Taguchi Approach", Australian Journal of Mechanical Engineering, Taylor and Francis. (**Under revision**).

***Journal Publications:***

- Panchanand Jha and B B Biswal, "A Neural Network Approach for Inverse Kinematic of a SCARA Manipulator", International Journal of Robotics and Automation (IJRA) Vol. 3, No. 1, March 2014, pp. 31~40, ISSN: 2089-4856.

- Panchanand Jha, Bibhuti B. Biswal, and Om Prakash Sahu, "Inverse Kinematic Solution of Robot Manipulator Using Hybrid Neural Network," International Journal of Materials Science and Engineering, Vol. 3, No. 1, pp. 31 -38, March 2015.

- Om Prakash Sahu, Bibhuti Bhusan Biswal, Saptarshi Mukharjee, and Panchanand Jha, "Development of Robotic End-Effector Using Sensors for Part Recognition and Grasping," International Journal of Materials Science and Engineering, Vol. 3, No. 1, pp. 39-43, March 2015.

- Panchanand Jha et al, "Development of an Intelligent System for Prediction of Inverse Kinematics of Robot Manipulator Arm ", IJNNA, International Science Press 5(2) July-December 2012, pp. 81 - 88.

- Panchanand Jha and B B Biswal, "Inverse Kinematic Solution of 5R Manipulator using ANN and ANFIS", International Journal of Robotics and Automation (IJRA) Vol. 4, No. 2, June 2015, ISSN: 2089-4856.

***Books Published:***

- Panchanand Jha and B B Biswal, "Kinematics ***with artificial neural network: A basic approach",*** *LAP LAMBERT Academic Publishing (2020-03-15), ISBN 978-620-2-51271-8.*

- *Panchanand Jha and B B biswal,* ***"OpenCV with Python****", Notion Press, ISBN No. 1685388086, 08/2021.*

***Book Chapters:***

- Panchanand Jha and Abinash Tripathy, "Prediction of diseases using machine-learning techniques", Computational Intelligence in Medical Decision Making and Diagnosis: Techniques and Applications", CRC Press, Taylor & Francis, 2023. Pages: 1-19, DOI: 10.1201/9781003309451-1

- Panchanand Jha and B B Biswal, "Intelligent Computation and Kinematics of 4-DOF SCARA Manipulator using ANN and ANFIS", Proceedings of Springer-Verlag Berlin Heidelberg, SEMCCO 2013, Part II, LNCS 8298.

- Panchanand Jha, Bibhuti B. Biswal, and Om Prakash Sahu, "Intelligent Computation of Inverse Kinematics of a 5-dof Manipulator Using MLPNN," Proceedings of Springer-Verlag Berlin Heidelberg, TAROS 2014, LNAI 8717, pp. 243–250, 2014.

- Panchanand Jha and B B Biswal, "Optimization Approach for Inverse Kinematic Solution", "Kinematics", book edited by Efren Gorrostieta Hurtado, ISBN 978-953-51-3688-0, Print ISBN 978-953-51-3687-3, Published: December 20, 2017.

***International Conferences:***

- Panchanand Jha and B B Biswal, "Inverse Kinematic Solution of Redundant Robot Manipulator using Artificial Neural Network ", Proceedings of International Conference on Advances in Mechanical Engineering May 29-31, 2013, COEP, Pune.

- Panchanand Jha and B B Biswal, "Development of an Intelligent System for Prediction of Inverse Kinematics of Robot Manipulator", 4th International & 25th All India Manufacturing Technology, Design and Research Conference (AIMTDR 2012) December 14th-16th, 2012, Jadavpur University, Kolkata, India.

- Bunil Kumar Balabantaray, Panchanand Jha and B B Biswal, "Application of Edge Detection Algorithm for Vision Guided Robotics Assembly System", Proceeding of SCIEI, the 6th International Conference on Machine Vision (ICMV 2013) London, the United Kingdom November 16-17, 2013.

- B B Biswal et, al., "Generation of Optimized Robotic Assembly Sequence using Immune Based Technique", Proceedings of the ASME 2012 International Mechanical Engineering Congress & Exposition IMECE2012 November 9-15, 2012, Houston, Texas, USA.

- Panchanand Jha and B B Biswal, "Hybrid training of neural network for the computation of inverse kinematics of revolute manipulator", AIMTDR, IIT Guwahati.

***National Conference:***

- Panchanand Jha, Pallavi Chandrakar and Shyam S. Dewangan, "Energy optimization and inverse kinematic solution of robot manipulator ", National conference on future clean and sustainable energy technologies, CCET, Bhilai, 2016

## Development programs and workshops

- Completed 12 weeks online NPTEL course on Strength of Materials with elite certification.
- Completed 4 weeks online NPTEL course on Ergonomics in the Automotive Design with elite certification.
- Completed 8 weeks online NPTEL course on kinematics of mechanisms and machine elite certification.
- Organized one-day National Seminar in the department of Mechanical Engineering, REC Visakhapatnam in 2017.
- Organized Workshop in NDT (Non-Destructive Testing) in REC, Visakhapatnam, 2018
- Online Webinar on ""Disassembly Planning of End-of-Life products - Environmental and Economic aspects" on 13/06/2020.
- 5 days online FDP on "Emerging technologies in robotics", 26-30 May 2020.
- 5 days online AICTE ATAL FDP on "Introduction to quantum computing and Information", IIITDM Jabalpur, July-2021
- 5 days online AICTE ATAL FDP on "Industrial Robots for future factories", NIT Puducherry, Sep-2021.
- 5-days AICTE ATAL FDP on REAL TIME PROBLEMS AND SOLUTIONS WITH AI & ML USING R PROGRAMMING.

- 5-days AICTE ATAL FDP on Microsoft and SAP Tech Saksham FDP on Artificial Intelligence by Techsaksham
- Workshop on Artificial Intelligence and Machine Learning Application in Healthcare organized by the Department of Computer Science and Engineering at NIT Meghalaya during 03rd -07th Sept. 2020.

## Prices, Medals, Awards and Honors received

- Research Fellowship from Ministry of Education, Science, Research and Sport of the Slovak Republic in 2014
- Research Fellowship Received CICOPS - Centre for International Cooperation and Development University of Pavia, Italy, 2017
- Ministry of Human Resource Development (Ministry of Education, India), Fellowship for higher education.

## Languages

- Hindi - Mother tongue
- English - Business Fluent